OHM'S LAW 1

In this worksheet you will learn how to use Ohm's law.

Georg Ohm was a German physicist (1787–1854). In 1826 he experimented and discovered that some conductors will keep a constant resistance value, no matter what the voltage applied to them. These devices are called ohmic conductors. Some things, like light globes and diodes, are called non-ohmic because their resistance changes as they get hotter.

special words

Ohm's law
voltage (V)
current (I)
resistance (R)
ohmic conductor
non-ohmic

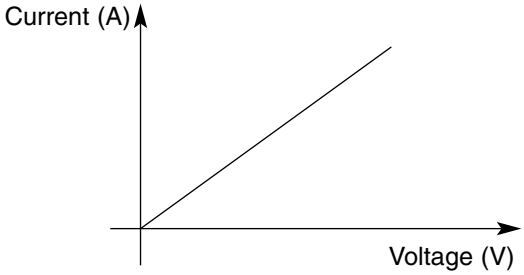


Ohm's law states that the resistance of some resistors is constant.

This can be written:

$$R = \frac{V}{I}$$

where R = resistance of conductor (Ω)
V = voltage across the conductor (V)
I = current through conductor (A)



We can rewrite Ohm's law in terms of voltage:

$$V = IR$$

Or in terms of current:

$$I = \frac{V}{R}$$

The graph above shows a straight line because the conductor has constant resistance. This is an ohmic conductor.

Examples

1 An electric drill is connected to a 240 V supply. It has a current of 3 A flowing through it. What is its resistance?

Step (i) Write out the information you are given: $V = 240$ V
$I = 3$ A

Step (ii) Choose which of the three formulas to use: $R = \frac{V}{I}$

Step (iii) Calculate the answer: $= \frac{240}{3}$
$= 80\ \Omega$

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2 A battery-powered radio has a resistance of 80 Ω and draws a current of 0.15 A. What is the total voltage of the batteries?

$R = 80\ \Omega$
$I = 0.15\ \text{A}$

Using $V = I \times R$
$= 80 \times 0.15$
$= 12\ \text{V}$

3 A vacuum cleaner plugged into a 240 V supply has a resistance of 60 Ω. What current will flow through it as it operates?

$V = 240\ \text{V}$
$R = 60\ \Omega$

Using $I = \frac{V}{R}$
$= \frac{240}{60}$
$= 4\ \text{A}$

To do



1 A fan heater draws a current of 10 A when operating on 240 V. What is its resistance?

(a) Write down your formula: $R =$

(b) Calculate the answer: _____

_____

_____

2 A refrigerator runs on mains voltage of 240 V and has a resistance of 400 Ω. What current does it draw?

(a) Write down the formula: $I =$

(b) Calculate the answer: _____

_____

_____

3 A portable tape recorder operates using a current of 1 A and has a resistance of 6 Ω. What voltage must the batteries be?

(a) Write down the formula: $V =$

(b) Calculate the answer: _____

_____

_____

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